

Vaibhav Chhaya

PhD Candidate, Department of Biology
University of Washington, Seattle

e-mail: chhaya@uw.edu
mobile: (206)751-5693

EDUCATION

PhD Candidate

Department of Biology, University of Washington (Seattle)
Advisor: Prof. Sharlene Santana

Sept. 2022 – present

Bachelor of Science-Master of Science (BS-MS Dual Degree)

Indian Institute of Science Education and Research, Pune
Advisor: Prof. Anand Krishnan

Aug. 2017 – May 2022

PEER-REVIEWED PUBLICATIONS

Santana S.E., Bernal-Rivera A., **Chhaya V.**, Guerra E., Quinche L.L., Visconti F., and Law C.J. (2025). An integrative perspective on bat evolution. *Annual Review of Ecology, Evolution, and Systematics*, 56.

King, N., Loeffler, J., Rosenberger, V., Wagner, F., **Chhaya, V.**, DeLeon, V., Curtis, A., Eiting, T. P., Santana, S. E., and Smith, T. D. (2025). Functional microanatomy of nasal turbinals in bats. *The Anatomical Record*, 1–60.([link](#))

Chhaya, V., Reddy, S. and Krishnan, A. (2023). Bill shape imposes biomechanical tradeoffs in cavity-excavating birds. *Proceedings of the Royal Society B*, 290, 1995.([link](#))

Mohan, R., **Chhaya, V.**, Krishnan, A. (2022). Seasonality and interspecific temporal partitioning in a semiarid grassland bat assemblage of northwestern India. *Journal of Arid Environments*, 205, 104818 ([link](#))

Chhaya, V.*, Lahiri, S.*, Jagan, M. A., Mohan, R., Pathaw, N. A., and Krishnan, A. (2021). Community Bioacoustics: Studying acoustic community structure for ecological and conservation insights. *Frontiers in Ecology and Evolution*, 9, 488. *equal contribution ([link](#))

BOOK CHAPTERS

Chhaya, V., Villalobos, D., (submitted). Darien Harvest Mouse *Reithrodontomys darienensis* (Pearson, 1939). In: León-Paniagua, L., Hernández-Canchola, G., Zeballos, H., Pardiñas, U.F. (eds) *Mammals of Middle and South America: Rodentia I*. Springer, Cham.

Villalobos, D., **Chhaya, V.**, (submitted). Chiriquian Harvest Mouse *Reithrodontomys garichensis* (Enders and Pearson, 1940). In: León-Paniagua, L., Hernández-Canchola, G., Zeballos, H., Pardiñas, U.F. (eds) *Mammals of Middle and South America: Rodentia I*. Springer, Cham.

Villalobos, D., **Chhaya, V.**, (submitted). Talamancan Harvest Mouse *Reithrodontomys creper* (Bangs, 1902). In: León-Paniagua, L., Hernández-Canchola, G., Zeballos, H., Pardiñas, U.F. (eds) *Mammals of Middle and South America: Rodentia I*. Springer, Cham.

Villalobos, D., **Chhaya, V.**, (submitted). Tacarcunan Bat *Lasiurus castaneus* (Handley, 1960.) In: Lim, B.K., López-González, C. (eds) *Mammals of Middle and South America: Chiroptera*. Springer, Cham.

Balebail, S., **Chhaya, V.**, Veith, J., Sisneros, J.A. (2026). Using Finite Element Modeling to Understand How Fish Resolve the 180° Ambiguity in Directional Hearing. In: Popper, A.N., Sisneros, J.A., Lepper, P.A., Vigness-Raposa, K.J. (eds) *The Effects of Noise on Aquatic Life IV*. Springer, Cham.

RESEARCH GRANTS

- The Society for Integrative and Comparative Biology Grant-in-Aid of Research (GIAR) (December 2025)
- The Company of Biologists Travelling Fellowship (December 2025)
- Jim and Nancy Kenagy Award, Department of Biology, University of Washington-Seattle (March 2025)
- Riddiford-Truman Award, Department of Biology, University of Washington-Seattle (March 2025)
- Margo and Tom Wyckoff Award, Department of Biology, University of Washington-Seattle (March 2024)
- Kishore Vaigyanik Protsahan Yojana (KVPY) Scholarship, Department of Science and Technology, Govt. of India (July 2017- July 2022)

HONORS AND AWARDS

- Mimi A.R. Koehl and Steven Wainwright Best Student Talk Award, Society for Integrative and Comparative Biology (SICB) Annual Meeting, Portland (January 2026)
- Ingrith-Deyrup Olsen Teaching Award, Department of Biology, University of Washington-Seattle (December 2025)
- Best Student Presentation (Runner-up Award), International Bat Research Conference (IBRC), Cairns (August 2025)
- Titley Scientific Award, North American Society for Bat Research (NASBR) Annual Conference, Guadalajara (October 2024)
- WRF Hall International Endowed Fellowship, Department of Biology, University of Washington (Seattle)
- Xytel Best MS Thesis Award for Biology, IISER Pune (May 2022)

CONFERENCE PRESENTATIONS

- Oral presentation on 'Convergent anatomical mechanisms of harmonic filtering in bat echolocation', **SICB Annual Meeting, Portland (January 2026)**
- Oral presentation on 'Echolocation shapes nasal morphology in bats', **International Bat Research Conference (IBRC), Cairns, Australia (August 2025)**
- Oral presentation on 'Functional interactions in the bat rostrum', **NASBR Annual Meeting, Guadalajara, Mexico (October 2024)**
- Poster on 'The effect of multifunctionality on morphological diversity in the bat rostrum', **SICB Annual Meeting, Seattle, USA (January 2024)**
- Virtual poster on 'The drivers and biomechanical consequences of bill shape diversification in Asian and African barbets', **SICB+ Annual Meeting (January - March 2022)**
- Oral presentation on 'Examining the patterns and drivers of bill shape evolution in Asian and African Barbets', **American Ornithological Society Annual Meeting (August 2021)**
- Poster presentation on 'Directional patterns in the song amplitude of Wren-Warblers', **National Symposium on Avian Biology, IISER Tirupati, India (December 2019)**

INVITED SEMINARS

- Biology department seminar, New Mexico State University, Las Cruces (January 2026)
- Pacific Northwest Acoustics Webinar, Applied Physics Laboratory, University of Washington (March 2026)',

TEACHING EXPERIENCE

- Instructor of Record at UW Biology for BIO497: Functional Morphology and Biomechanics Seminar (Fall 2024)
- Teaching assistant at UW Biology for BIO427: Biomechanics (Winter 2024 and 2025), BIO444: Ornithology (Spring 2024), BIO220: Introductory Biology (Fall 2023)

MENTORSHIP EXPERIENCE

- Sophie Garrote (Undergraduate, School of Environmental and Forest Sciences, University of Washington): Guiding an independent research project on the evolution of novel resonators in the bat nasopharynx, including project design, data analysis, and scientific communication. Sophie was recently selected for the Undergraduate Community at Evolution (UCE) program by the Society for the Study of Evolution.
- Derek Young (Research Volunteer, Department of Biology, University of Washington): Supervising a project on patterns of diversification in the bat vomeronasal organ, with training in comparative morphology and data analysis.

OTHER PROFESSIONAL SERVICE

- Graduate representative, UW Biology Conflict Resolution Sub-Committee (September 2024-25) .
- Graduate representative, UW Biology Diversity and Equity Committee (September 2023-25) .
- Volunteer at the Burke Museum of Natural History and Culture (University of Washington). Here, I have been involved with preparing and sorting specimens for the mammalogy collections.
- Volunteer at the Science Nurture Programme (August 2018 – July 2019), an initiative by IISER students to popularise science among students of Pune Municipal schools. Here, I designed and taught sessions on diverse topics in biology and natural history using hands-on activities and field trips.

OUTREACH AND MEDIA

- Popularising bat research and conservation at the Burke Member's Night (February 2024) and the Burke Rare Air Reception (March 2025)
- Interviewed for a Deccan Herald article covering published research on cavity nest excavation in barbets ([link](#))
- Interview for Microphotonics 'Image of the Month' covering research on bat rostrum evolution ([link](#))

AD HOC REVIEWER FOR

- Royal Society Open Science
- Proceedings of the Royal Society B
- Zoological Journal of the Linnean Society
- Integrative Organismal Biology
- Palaeontologia Electronica